Area and surface area

20 questions, including 2 challenges. Write your work and answers on separate paper.

1. 9×3

2. 9×7

3. 9×4

4. 9×9

5. Find $1{,}344 \div 32$. Verify by multiplication.

6. 9 of 12 tiles are green. What percent are green?

7. 3×5

8. A parallelogram has base 6, height 8, and slanted side 10 cm. Find its area.

9. A rectangle is 6 cm by $\frac{3}{2}$ cm. Find its area.

10. A parallelogram has base 9, height 4, and slanted side 5 cm. Find its area.

11. A triangle has base 9 cm and height 4 cm. Find its area.

12. A 9 by 5 cm rectangle loses a 2 by 3 cm corner. Find the remaining area.

13. Theo uses a triangle's base 9 cm and height 4 cm. He writes $9 \times 4 = 36$ square cm. What is missing?

14. A rectangle has length 8 cm and width 5 cm. Find area and perimeter.

15. A parallelogram has base 6 cm, height 5 cm, and slanted side 7 cm. Find area and name the unused length.

16. A right triangle has perpendicular legs 12 cm and 5 cm. Find area using either leg as base.

17. A triangle has area 30 square cm and base 12 cm. Find its matching height.

18. A9 cm by 7 cm rectangle loses a3 cm by 2 cm corner. Find remaining area.

CHALLENGE / Explain your plan and check the result

C1. A wall is 8 m wide and 3 m high. It contains a2 m by 1.5 m window and a triangular vent with base 2 m and height 1 m. Paint covers 5 square m per liter. How many whole liters must be bought?

C2. Two triangular flags each have area 24 square cm. Flag A has base 8 cm; flag B has base 6 cm. Find both heights. Which flag is taller, and by what factor? Explain using area, not a sketch's appearance.